Chapter 6 - Quadratic Equations

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2. Roots of a Quadratic Equation
3. **Proof**: The quadratic equation cannot have more than two roots.
4. Nature of Roots
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#Fundamental Theorem of Algebra

The fundamental theorem of algebra states that 'Every equation of degree n in x has n roots, and no more'.

#Roots of a Quadratic Equation

$$ax^2 + bx + c = 0 \Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\therefore x_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } x_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

#Nature of Roots

For $ax^2 + bx + c = 0$,

1. $(b^2 - 4ac > 0) \Rightarrow$ Roots are real and unequal.
    a. $b^2 - 4ac = n^2, (\exists n \in N) \Rightarrow$ Roots are rational.
    b. $b^2 - 4ac \neq n^2, (\exists n \in N) \Rightarrow$ Roots are irrational.
2. $(b^2 - 4ac = 0) \Rightarrow$ Roots are real and equal.
3. $(b^2 - 4ac < 0) \Rightarrow$ Roots are imaginary and unequal.

Discriminant

The quantity $(b^2 - 4ac)$ which determines the nature of the roots of a quadratic equation in x is called discriminant with respect to x.

#Relation Between Roots and Coefficients

If α and β are the roots of quadratic equation $ax^2 + bx + c = 0$ then

1. Sum of Roots (S) $= \alpha + \beta = \frac{-b}{a}$
2. Product of Roots (P) $= \alpha\beta = \frac{c}{a}$

#Special Roots Condition

For $ax^2 + bx + c = 0$,

1. Roots equal in magnitude but opposite in sign: $b = 0$
2. Reciprocal roots: $c = a$
3. One root zero: $c = 0$
4. Both roots zero: $b = 0$ and $c = 0$

#Formation of a Quadratic Equation

If S and P are the sum of roots and product of roots respectively then the quadratic equation is

$x^2 - Sx + P = 0$

#Common Root Condition

If two quadratic equations are $ax^2 + bx + c = 0$ and $a'x^2 + b'x + c' = 0$ then

1. One root common:

   $(ab' - a'b)(bc' - b'c) = (ca' - c'a)^2$

   or,

   $$\begin{vmatrix} a & b \\ a' & b' \end{vmatrix} \begin{vmatrix} b & c \\ b' & c' \end{vmatrix} = \begin{vmatrix} c & a \\ c' & a' \end{vmatrix}^2$$

2. Both roots common:

   $$\frac{a}{a'} = \frac{b}{b'} = \frac{c}{c'}$$